

ASTGHİK HAKOBYAN, Ph.D.

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Google Scholar • Citations: 493 • h-index: 10 • ORCID

Research Summary. My research focuses on safe autonomy and decision-making under uncertainty for robotic and autonomous systems. I develop distributionally robust optimization and risk-aware control methods that enable learning-enabled systems to operate reliably despite uncertain and imperfect information.

Research Interests: Safe Autonomy • Distributionally Robust Control • Motion Planning • Autonomous Systems • Optimization

Professional Experience

Assistant Professor 2026–Present
Zaven & Sonia Akian College of Science and Engineering (ACSE), American University of Armenia, Yerevan

Head 2026–Present
“Intelligent Systems and Advanced Control” Basic Research Laboratory, National Polytechnic University of Armenia, Yerevan

Leading Researcher 2023–Present
Center for Scientific Innovation and Education (CSIE), Yerevan

Assistant Professor, Control Systems 2023–2025
National Polytechnic University of Armenia, Yerevan

Researcher Sep 2020–Feb 2021
Automation and Systems Control Research Institute,
Seoul National University, Seoul

Application Engineer 2017–2018
National Instruments AM LLC, Yerevan

Education

Ph.D., Electrical and Computer Engineering 2023
Seoul National University, Seoul, Korea
Thesis: Wasserstein Distributionally Robust Control and Optimization for Autonomous Systems
(*Distinguished ECE Ph.D. Dissertation Award*)
Advisor: Prof. Insoon Yang

M.S., Electrical and Computer Engineering 2020
Seoul National University, Seoul, Korea
Thesis: Risk-Aware Distributionally Robust Optimization for Learning-Based Autonomous Systems

(Distinguished ECE M.S. Dissertation Award)

Advisor: Prof. Insoon Yang

B.S., Automation and Control

2018

National Polytechnic University of Armenia, Yerevan

Final Project: Design and Analysis of SISO/MIMO Hydraulic Control Systems

Advisor: Prof. Azatuhi Ulikyan

Research Grants

Principal Investigator — Incoming Research Fellowship Program (2024–2029)

Higher Education and Science Committee of the RA MoESCS

Grant ID: 24IRF-2B002

Principal Investigator — Research Projects in Prospective Directions (2024–2027)

Higher Education and Science Committee of the RA MoESCS

Grant ID: 24FP-2C017

Publications

Peer-Reviewed Journal Articles

- [1] A. Hakobyan and I. Yang, “Wasserstein distributionally robust control of partially observable linear stochastic systems,” *IEEE Transactions on Automatic Control*, 2024.
- [2] J. M. Nadasles, A. Hakobyan, D. M. de la Peña, D. Limon, and I. Yang, “Risk-aware Wasserstein distributionally robust control of vessels in natural waterways,” *IEEE Transactions on Control Systems Technology*, vol. 32, no. 4, pp. 1471–1478, 2024.
- [3] A. Hakobyan and I. Yang, “Distributionally robust differential dynamic programming with Wasserstein distance,” *IEEE Control Systems Letters*, vol. 7, pp. 2329–2334, 2023, *Presented at CDC 2023*.
- [4] A. Hakobyan and I. Yang, “Distributionally robust risk map for learning-based motion planning and control: A semidefinite programming approach,” *IEEE Transactions on Robotics (T-RO)*, 2022.
- [5] J. Shin, A. Hakobyan, M. Park, Y. Kim, G. Kim, and I. Yang, “Infusing model predictive control into meta-reinforcement learning for mobile robots in dynamic environments,” *IEEE Robotics and Automation Letters (RA-L)*, pp. 1–8, 2022, *Presented at IROS 2022*.
- [6] A. Hakobyan and I. Yang, “Wasserstein distributionally robust motion control for collision avoidance using conditional value-at-risk,” *IEEE Transactions on Robotics (T-RO)*, vol. 38, no. 2, pp. 939–957, 2021.
- [7] A. Hakobyan, G. C. Kim, and I. Yang, “Risk-aware motion planning and control using CVaR-constrained optimization,” *IEEE Robotics and Automation Letters (RA-L)*, vol. 4, no. 4, pp. 3924–3931, 2019, *Presented at IROS 2019*.

Conference Papers

- [1] M. Jang, A. Hakobyan, and I. Yang, “On the steady-state distributionally robust Kalman filter,” in *2025 IEEE 64th Conference on Decision and Control (CDC)*, 2025, pp. 2385–2392.
- [2] A. Hakobyan and I. Yang, “Distributionally robust optimization with unscented transform for learning-based motion control in dynamic environments,” in *IEEE International Conference on*

Robotics and Automation (ICRA), 2023, pp. 3225–3232.

- [3] A. Hakobyan and I. Yang, “Wasserstein distributionally robust control of partially observable linear systems: Tractable approximation and performance guarantee,” in *IEEE Conference on Decision and Control (CDC)*, 2022, pp. 4800–4807.
- [4] A. Hakobyan and I. Yang, “Toward improving the distributional robustness of risk-aware controllers in learning-enabled environments,” in *IEEE Conference on Decision and Control (CDC)*, 2021, pp. 6024–6031.
- [5] A. Hakobyan and I. Yang, “Learning-based distributionally robust motion control with Gaussian processes,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2020, pp. 7667–7674.
- [6] A. Hakobyan and I. Yang, “Wasserstein distributionally robust motion planning and control with safety constraints using conditional value-at-risk,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 490–496.

Book Chapter

- [1] A. Cherukuri, A. Dixit, and A. Hakobyan, *Distributionally Robust and Risk-Averse MPC for Motion Planning and Control: Reformulations and Computational Issues*. Cham: Springer International Publishing, 2025.

Preprints

- [1] A. Hakobyan, A. Nazarians, A. Gahlawat, N. Hovakimyan, and I. Kolmanovsky, “Distributionally robust planning with $\mathcal{L}_1$ adaptive control,” *arXiv preprint arXiv:2603.28758*, 2026, *accepted for presentation at CDC 2026*.
- [2] M. Jang, J. Lee, A. Hakobyan, N. Hovakimyan, and I. Yang, “Residual-aware distributionally robust EKF: Absorbing linearization mismatch via wasserstein ambiguity,” *arXiv preprint arXiv:2604.02749*, 2026, *accepted for presentation at CDC 2026*.
- [3] M. Jang, A. Hakobyan, and I. Yang, “Distributionally robust Kalman filter,” *arXiv preprint arXiv:2512.06286*, 2025, *Under review for TAC*.
- [4] M. Jang, A. Hakobyan, and I. Yang, “Wasserstein distributionally robust control and state estimation for partially observable linear systems,” *arXiv preprint arXiv:2406.01723*, 2024.

Teaching Experience

Instructor, ARCS.ai (Fall 2025)

- Introduction to Robot Control

Instructor, National Polytechnic University of Armenia

- Robot Motion Planning and Control (Spring 2025)
- Optimal Control Theory (Spring 2025)
- Estimation Theory (Fall 2024)
- Industrial Control Systems (Fall 2023)

Teaching Assistant, Seoul National University

- Advanced Control Techniques (Fall 2022, Fall 2021)
- Feedback Control Systems (Fall 2019)
- Introduction to Robotics and Autonomous Systems (Spring 2019)

Student Supervision

M.Sc. Students

- 1 student supervised (graduated 2025), National Polytechnic University of Armenia
- 1 student supervised (expected 2026), National Academy of Sciences of the Republic of Armenia

Undergraduate Students

- 2 students supervised (graduated 2025), National Polytechnic University of Armenia
- 1 student supervised (expected 2026), American University of Armenia

CSIE Interns

- Supervised over 10 research interns at CSIE, several of whom are currently enrolled in graduate programs in Armenia and abroad.

Conference & Workshop Organization

Workshop Organizer — Safety as a System Property: From Control Design to Data-Driven Autonomy, 23rd IFAC World Congress, 2026

Workshop Organizer — Safe Control and Learning in Autonomous Systems, European Control Conference (ECC), 2025

Workshop Organizer — Innovations in Autonomous Systems, World Congress on Innovation and Technology (WCIT), 2024

Professional Service

Program Committee Member — IEEE FINE 2026 (International Conference on Future Networks and Intelligence)

Technical Committee Member — IEEE Control Systems Society, Aerospace Control Technical Committee

National Scientific Service — Subcommittee and Review Board Member, Higher Education and Science Committee (HESC), Ministry of Education, Science, Culture and Sports of the Republic of Armenia

Professional Memberships — IEEE, American Association for the Advancement of Science (AAAS), Armenian Society of Fellows (ASOF)

Awards and Honors

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| • Distinguished ECE Ph.D. Dissertation Award | 2023 |
| • SNU Global Scholarship | 2022 |
| • SNU Development Fund Tuition Scholarship | 2021 |
| • Distinguished ECE M.S. Dissertation Award | 2020 |
| • Korean Government Scholarship Program | 2018–2020 |
| • Armenian Government Academic Scholarship for Excellence | 2015–2018 |
| • Knights of Vartan Avak Tahlij Scholarship Award | 2016 |

Languages

- Armenian (Native)
- Russian (Fluent)
- English (Fluent, TOEFL iBT 107)
- Korean (Fluent, TOPIK 6)
- Chinese (Moderate, HSK 3)